

# Database Laboratory Work 4

SQL Queries and Results  
PostgreSQL / pgAdmin

# 1. Uppercase airline names

Retrieves all airline names in uppercase.

```
SELECT UPPER(name) AS airline_name
FROM airline;
```

Task 1 • Database Laboratory Work 4

The screenshot displays the pgAdmin 4 web interface. On the left, the 'Servers' tree is expanded to show the 'airline\_db' database. The main pane shows a SQL query: `SELECT UPPER(name) AS airline_name FROM airline;`. Below the query, the 'Data Output' tab is active, showing a table with 5 rows of results. The table has a single column named 'airline\_name' of type 'text'. The results are: KAZAIR, AIREASY, FLYHIGH, FLYFLY, and GLOBAL AIRWA... (truncated). The interface also includes a 'Query History' tab and a 'Scratch Pad' tab. The bottom status bar indicates 'Showing rows: 1 to 5' and 'Page No: 1 of 1'.

| airline_name      |
|-------------------|
| 1 KAZAIR          |
| 2 AIREASY         |
| 3 FLYHIGH         |
| 4 FLYFLY          |
| 5 GLOBAL AIRWA... |

Активация  
Чтобы активировать  
раздел "Параметры"

## 2. Replace Air with Aero

Replaces every occurrence of “Air” in airline names with “Aero”.

```
SELECT name,  
REPLACE(name, 'Air', 'Aero') AS  
new_name  
FROM airline;
```

Task 2 • Database Laboratory Work 4

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer shows the database structure: Servers (4) > PostgreSQL 18 > Databases (3) > airport\_db. The main pane displays a SQL query:

```
1 SELECT name,  
2     REPLACE(name, 'Air', 'Aero') AS new_name  
3 FROM airline;
```

The Data Output pane shows the results of the query:

|   | name           | new_name         |
|---|----------------|------------------|
| 1 | KazAir         | KazAero          |
| 2 | AirEasy        | AeroEasy         |
| 3 | FlyHigh        | FlyHigh          |
| 4 | FlyFly         | FlyFly           |
| 5 | Global Airways | Global Aerowa... |

The status bar at the bottom indicates "Total rows: 5" and "Query complete 00:00:00.120".

### 3. Flights for airline 1 and 2

Finds flight IDs belonging to airline 1 or airline 2. The schema has

```
SELECT flight_id
FROM flight
WHERE airline_id IN (1, 2);
```

Task 3 • Database Laboratory Work 4

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer shows the database structure: Servers (4) > PostgreSQL 15 > PostgreSQL 16 > PostgreSQL 18 > Databases (3) > airport\_db > flight. The main pane shows the SQL query:

```
SELECT flight_id
FROM flight
WHERE airline_id IN (1, 2);
```

The Data Output pane shows the results of the query:

| flight_id | [PK] integer |
|-----------|--------------|
| 1         | 5            |

The bottom status bar shows "Total rows: 1" and "Query complete 00:00:00.142".

Активация Windows  
Чтобы активировать Windows, перейдите в раздел "Параметры".

23:23:01  
05.10.2026

## 4. Airports containing Reginal and Air

Searches airport names containing both requested words. Result: 0 rows in the current database.

```
SELECT *  
FROM airport  
WHERE airport_name ILIKE '%Reginal%'  
AND airport_name ILIKE '%Air%';
```

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including the 'airport\_db' database. The main query editor on the right contains the following SQL query:

```
1 SELECT *  
2 FROM airport  
3 WHERE airport_name ILIKE '%Reginal%'  
4 AND airport_name ILIKE '%Air%';
```

Below the query editor, the Data Output tab shows the results of the query. The table has the following columns: airport\_id (integer), airport\_name (character varying (100)), country (character varying (100)), state (character varying (100)), city (character varying (100)), created\_at (timestamp without time zone), and updated\_at (timestamp without time zone). The results show 0 rows.

At the bottom of the interface, a Windows activation watermark is visible, stating: "Активация Windows. Чтобы активировать Windows, перейдите в раздел 'Параметры'." The system tray at the bottom shows the date and time as 23:27:06 on 05.10.2026.

# 5. Passenger names and birth dates

Displays passenger names and formats dates as Month DD, YYYY.

```
SELECT
    first_name,
    last_name,
    TO_CHAR(date_of_birth, 'FMMonth DD,
    YYYY') AS birth_date
FROM passenger;
```

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer shows the database structure, with 'airport\_db' selected. The main pane displays a SQL query in the Query editor:

```
1 SELECT
2     first_name,
3     last_name,
4     TO_CHAR(date_of_birth, 'FMMonth DD, YYYY') AS birth_date
5 FROM passenger;
```

Below the query editor, the Data Output pane shows the results of the query in a table format:

|   | first_name<br>character varying (50) | last_name<br>character varying (50) | birth_date<br>text  |
|---|--------------------------------------|-------------------------------------|---------------------|
| 1 | Ayan                                 | Saparov                             | January 15, 2000    |
| 2 | Aruzhan                              | Bekova                              | May 20, 2001        |
| 3 | Dias                                 | Nurgali                             | September 10, 19... |

The bottom status bar indicates 'Total rows: 3' and 'Query complete 00:00:00.089'. A Windows activation watermark is visible in the bottom right corner.

## 6. Delayed flights

Finds flights whose actual arrival was later than the scheduled arrival.

```
SELECT flight_id,  
       scheduled_arrival,  
       actual_arrival  
FROM flight  
WHERE actual_arrival >  
       scheduled_arrival;
```

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including Servers (4), Databases (3), and the selected database 'airport\_db' with its various components like Casts, Catalogs, Event Triggers, Extensions, Foreign Data Wrappers, Languages (1), Publications, Schemas, Subscriptions, postgres, postgres\_db, Login/Group Roles, Tablespaces, phonebook, and db.

The main pane shows a SQL query in the 'Query' tab:

```
1 SELECT flight_id,  
2       scheduled_arrival,  
3       actual_arrival  
4 FROM flight  
5 WHERE actual_arrival > scheduled_arrival;
```

The 'Data Output' tab at the bottom displays the results of the query in a table with 3 rows and 3 columns: flight\_id, scheduled\_arrival, and actual\_arrival.

|   | flight_id<br>[PK] integer | scheduled_arrival<br>timestamp without time zone | actual_arrival<br>timestamp without time zone |
|---|---------------------------|--------------------------------------------------|-----------------------------------------------|
| 1 | 3                         | 2023-06-20 15:00:00                              | 2023-06-20 15:05:00                           |
| 2 | 4                         | 2025-03-12 16:00:00                              | 2025-03-12 16:05:00                           |
| 3 | 5                         | 2022-11-01 11:00:00                              | 2022-11-01 11:05:00                           |

The status bar at the bottom indicates 'Total rows: 3' and 'Query complete 00:00:00.092'. The system tray at the bottom right shows the date and time: 23:50:12 05.10.2026.

# 7. Flights that arrived late

Shows the flight ID and both arrival times for late flights.

```
SELECT flight_id,  
       scheduled_arrival,  
       actual_arrival  
FROM flight  
WHERE actual_arrival >  
       scheduled_arrival;
```

Task 7 • Database Laboratory Work 4

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer shows the database structure, including Servers (4), PostgreSQL 15, PostgreSQL 16, PostgreSQL 18, and Databases (3). The 'airport\_db' database is selected, showing its contents: Casts, Catalogs, Event Triggers, Extensions, Foreign Data Wrappers, Languages (1), Publications, Schemas, Subscriptions, postgres, postgres\_db, Login/Group Roles, Tablespace, phonebook, and db.

The main pane displays a SQL query in the Query editor:

```
1 SELECT  
2   flight_id,  
3   scheduled_arrival,  
4   actual_arrival  
5 FROM flight  
6 WHERE actual_arrival > scheduled_arrival;
```

The Query History pane shows the same query. The Data Output pane displays the results of the query:

|   | flight_id<br>(PK) integer | scheduled_arrival<br>timestamp without time zone | actual_arrival<br>timestamp without time zone |
|---|---------------------------|--------------------------------------------------|-----------------------------------------------|
| 1 | 3                         | 2023-06-20 15:00:00                              | 2023-06-20 15:05:00                           |
| 2 | 4                         | 2025-03-12 16:00:00                              | 2025-03-12 16:05:00                           |
| 3 | 5                         | 2022-11-01 11:00:00                              | 2022-11-01 11:05:00                           |

The bottom status bar shows: Total rows: 3, Query complete 00:00:00.085, CRLF, Ln 6, Col 42. A green notification box at the bottom right states: "Activation Windows. Чтобы активировать Windows, перейдите в раздел 'Параметры'. Successfully run. Total query runtime: 85 msec. 3 rows affected."

## 8. Airlines by country and date

Finds airlines from the specified countries created in the specified period. Result: 0 rows.

```
SELECT *  
FROM airline  
WHERE country IN ('France', 'Portugal',  
                  'Poland')  
AND created_at BETWEEN '2023-11-01'  
AND '2024-03-31';
```

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including Servers (4), PostgreSQL 15, PostgreSQL 16, PostgreSQL 18, and Databases (3). The 'airline\_db' database is selected, showing its contents: Casts, Catalogs, Event Triggers, Extensions, Foreign Data Wrappers, Languages (1), Publications, Schemas, Subscriptions, postgres, postgres\_db, Login/Group Roles, Tablespaces, phonebook, and db.

The main pane shows the SQL query editor with the following query:

```
1 SELECT *  
2 FROM airline  
3 WHERE country IN ('France', 'Portugal', 'Poland')  
4 AND created_at BETWEEN '2023-11-01' AND '2024-03-31';
```

The Data Output pane shows the results of the query, which are empty. The table structure is displayed as follows:

| airline_id   | airline_code           | name                    | country                 | created_at                  | updated_at                  |
|--------------|------------------------|-------------------------|-------------------------|-----------------------------|-----------------------------|
| [PK] integer | character varying (10) | character varying (100) | character varying (100) | timestamp without time zone | timestamp without time zone |

The status bar at the bottom indicates "Total rows: 0" and "Query complete 00:00:00.133".

Активация Windows  
Чтобы активировать Windows, перейдите в раздел "Параметры".

Task 8 • Database Laboratory Work 4

## 9. Top 3 overweight baggage

Finds the 3 heaviest baggage records over 25 kg. Current baggage table is empty, so result: 0 rows.

```
SELECT *  
FROM baggage  
WHERE weight_kg > 25  
ORDER BY weight_kg DESC  
LIMIT 3;
```

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including Servers (4), PostgreSQL 15, PostgreSQL 16, PostgreSQL 18, and Databases (3). The 'airport\_db' database is selected, showing its contents: Casts, Catalogs, Event Triggers, Extensions, Foreign Data Wrappers, Languages (1), Publications, Schemas, Subscriptions, postgres, postgres\_db, Login/Group Roles, Tablespaces, phonebook, and db.

The main pane shows the SQL query editor with the following query:

```
1 SELECT *  
2 FROM baggage  
3 WHERE weight_kg > 25  
4 ORDER BY weight_kg DESC  
5 LIMIT 3;
```

The Query History tab shows the same query. The Data Output tab displays the results of the query, which are empty (0 rows). The table structure is as follows:

| baggage_id   | booking_id | weight_kg     | created_at                  | updated_at                  |
|--------------|------------|---------------|-----------------------------|-----------------------------|
| [PK] integer | integer    | numeric (5,2) | timestamp without time zone | timestamp without time zone |

The status bar at the bottom indicates 'Total rows: 0' and 'Query complete 00:00:00.085'. The Windows taskbar at the bottom shows the system clock as 02:26:34 on 06.10.2026.

# 10. Youngest passenger

Returns the passenger with the latest date of birth, i.e. the youngest passenger.

```
SELECT first_name, last_name
FROM passenger
ORDER BY date_of_birth DESC
LIMIT 1;
```

The screenshot shows the pgAdmin 4 web interface. On the left, the 'Object Explorer' pane displays a tree structure of databases, with 'airport\_db' selected. The main pane shows a SQL query editor with the following query:

```
1 SELECT first_name, last_name
2 FROM passenger
3 ORDER BY date_of_birth DESC
4 LIMIT 1;
```

Below the query editor, the 'Data Output' pane displays the results of the query in a table format:

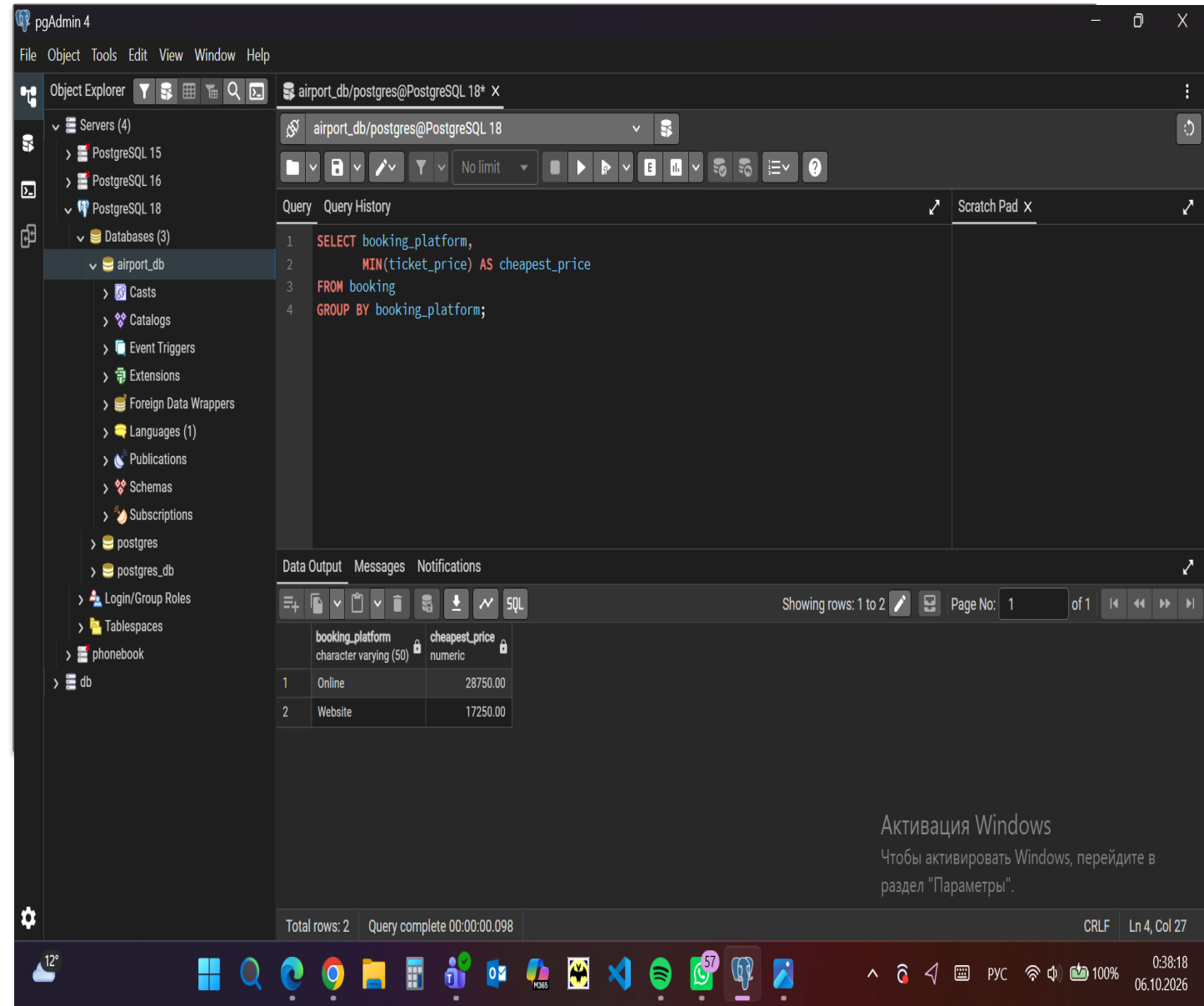
|   | first_name<br>character varying (50) | last_name<br>character varying (50) |
|---|--------------------------------------|-------------------------------------|
| 1 | Aruzhan                              | Bekova                              |

The bottom status bar indicates 'Total rows: 1' and 'Query complete 00:00:00.087'. A Windows activation watermark is visible in the bottom right corner of the screenshot.

# 11. Cheapest booking by platform

Finds the minimum ticket price for each booking platform.

```
SELECT booking_platform,  
       MIN(ticket_price) AS  
       cheapest_price  
FROM booking  
GROUP BY booking_platform;
```



The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer shows the database structure, with 'airport\_db' selected. The main pane displays a SQL query:

```
1 SELECT booking_platform,  
2       MIN(ticket_price) AS cheapest_price  
3 FROM booking  
4 GROUP BY booking_platform;
```

Below the query, the Data Output tab shows the results of the query:

|   | booking_platform<br>character varying (50) | cheapest_price<br>numeric |
|---|--------------------------------------------|---------------------------|
| 1 | Online                                     | 28750.00                  |
| 2 | Website                                    | 17250.00                  |

The status bar at the bottom indicates 'Total rows: 2' and 'Query complete 00:00:00.098'. A Windows activation watermark is visible in the bottom right corner.

## 12. Airline codes containing digits

Finds airlines whose airline code contains at least one digit. Result: 0 rows.

```
SELECT *  
FROM airline  
WHERE airline_code ~ '[0-9]';
```

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer shows the database structure, including the 'airline\_db' database. The main pane displays the query results for the query: `SELECT * FROM airline WHERE airline_code ~ '[0-9]';`. The results are shown in a table with columns: `airline_id` (integer), `airline_code` (character varying (10)), `name` (character varying (100)), `country` (character varying (100)), `created_at` (timestamp without time zone), and `updated_at` (timestamp without time zone). The table is empty, indicating 0 rows. The status bar at the bottom shows 'Total rows: 0' and 'Query complete 00:00:00.093'. A Windows activation watermark is visible in the bottom right corner.

| airline_id | airline_code | name | country | created_at | updated_at |
|------------|--------------|------|---------|------------|------------|
|------------|--------------|------|---------|------------|------------|

# 13. Top 5 newest airlines

Returns the 5 most recently created airline records.

```
SELECT *
FROM airline
ORDER BY created_at DESC
LIMIT 5;
```

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

Servers (4)

- PostgreSQL 15
- PostgreSQL 16
- PostgreSQL 18
  - Databases (3)
    - airport\_db
      - Casts
      - Catalogs
      - Event Triggers
      - Extensions
      - Foreign Data Wrappers
      - Languages (1)
      - Publications
      - Schemas
      - Subscriptions
    - postgres
    - postgres\_db
    - Login/Group Roles
    - Tablespaces
  - phonebook
  - db

airport\_db/postgres@PostgreSQL 18\* X

airport\_db/postgres@PostgreSQL 18

No limit

Query Query History

Scratch Pad X

1 SELECT \*

2 FROM airline

3 ORDER BY created\_at DESC

4 LIMIT 5;

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

|   | airline_id<br>[PK] integer | airline_code<br>character varying (10) | name<br>character varying (100) | country<br>character varying (100) | created_at<br>timestamp without time zone | updated_at<br>timestamp without time zone |
|---|----------------------------|----------------------------------------|---------------------------------|------------------------------------|-------------------------------------------|-------------------------------------------|
| 1 | 5                          | OLD                                    | Global Airways                  | United Kingdom                     | 2026-09-28 23:56:59.047291                | 2026-09-28 23:57:32.093141                |
| 2 | 2                          | AEE                                    | AirEasy                         | FRANCE                             | 2026-09-28 20:37:43.738358                | 2026-09-28 20:37:43.738358                |
| 3 | 3                          | FLH                                    | FlyHigh                         | BRAZIL                             | 2026-09-28 20:37:43.738358                | 2026-09-28 20:37:43.738358                |
| 4 | 4                          | FLY                                    | FlyFly                          | POLAND                             | 2026-09-28 20:37:43.738358                | 2026-09-28 20:37:43.738358                |
| 5 | 1                          | KAZ                                    | KazAir                          | TURKEY                             | 2026-09-28 20:35:45.529693                | 2026-09-28 20:35:45.529693                |

Total rows: 5 Query complete 00:00:00.091

CRLF Ln 4, Col 9

Активация Windows  
Чтобы активировать Windows, перейдите в раздел "Параметры".

# 14. Booking IDs 200–300, not Checked

Finds baggage checks for booking IDs 200–300 whose result is not Checked. Result: 0 rows.

```
SELECT *  
FROM baggage_check  
WHERE booking_id BETWEEN 200 AND 300  
AND check_result <> 'Checked';
```

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, with 'airport\_db' selected. The main pane shows a SQL query: `SELECT * FROM baggage_check WHERE booking_id BETWEEN 200 AND 300 AND check_result <> 'Checked';`. The Query History tab is active, showing the query. The Data Output tab is also visible, displaying the results of the query. The results table has columns: `baggage_check_id` (integer), `booking_id` (integer), `passenger_id` (integer), `check_result` (character varying (50)), `created_at` (timestamp without time zone), and `updated_at` (timestamp without time zone). The status bar at the bottom indicates 'Total rows: 0' and 'Query complete 00:00:00.085'. A Windows activation watermark is visible in the bottom right corner.

| baggage_check_id | booking_id | passenger_id | check_result | created_at | updated_at |
|------------------|------------|--------------|--------------|------------|------------|
|------------------|------------|--------------|--------------|------------|------------|

# 15. Updated earlier in the same month

Finds baggage checks updated earlier than creation time while remaining in the same month. Result: 0 rows.

```
SELECT *  
FROM baggage_check  
WHERE DATE_TRUNC('month', updated_at) =  
DATE_TRUNC('month', created_at)  
AND updated_at < created_at;
```

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays a tree view of the database structure, including Servers (4), PostgreSQL 15, PostgreSQL 16, PostgreSQL 18, and Databases (3). The 'airport\_db' database is selected, showing its contents: Casts, Catalogs, Event Triggers, Extensions, Foreign Data Wrappers, Languages (1), Publications, Schemas, Subscriptions, postgres, postgres\_db, Login/Group Roles, Tablespaces, phonebook, and db.

The main pane shows the SQL query editor with the following query:

```
1 SELECT *  
2 FROM baggage_check  
3 WHERE DATE_TRUNC('month', updated_at) = DATE_TRUNC('month', created_at)  
4 AND updated_at < created_at;
```

The Query History pane shows the query was executed successfully. The Data Output pane displays the results of the query, showing columns: baggage\_check\_id (integer), booking\_id (integer), passenger\_id (integer), check\_result (character varying (50)), created\_at (timestamp without time zone), and updated\_at (timestamp without time zone). The results show 0 rows.

The bottom status bar indicates: Total rows: 0, Query complete 00:00:00.094, CRLF, Ln 4, Col 31.

A Windows activation watermark is visible in the bottom right corner of the screenshot.

# Conclusion

Laboratory Work 4 completed using PostgreSQL / pgAdmin.

**15 SQL tasks completed**

Screenshots of SQL code and results are included on the task slides.